

Microsoft®
tech·ed
North America | 2010

JUNE 7-10, 2010 | NEW ORLEANS, LA



Microsoft®



Let Build Automation in Microsoft Visual Studio Team Foundation Server 2010 Work for You

Richard Hundhausen

President, Accentient, Inc.

Visual Studio ALM MVP



Richard Hundhausen



richard@accentient.com



[@rhundhausen](https://twitter.com/rhundhausen)

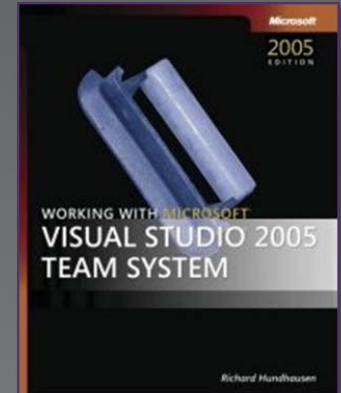


<http://www.accentient.com>



<http://blog.accentient.com>

Microsoft®
Regional Director
PROGRAM



Agenda

- Overview
- Configuration
- Build definitions
- WF or MSBuild?
- Gated Check-Ins
- Private Builds
- Customization
- FAQs
- Demos



Overview

- Team Foundation Build is a core part of Team Foundation Server
- There are no extra components to buy or license
- Many testing and analysis components are built-in
- TFS 2010 build service can be installed on a TFS 2008 build machine
 - They both default to port 9191 but they are capable of sharing it



What's New in Team Foundation Build 2010

- Build Controller and Build Agent architecture
- Windows Workflow 4.0 based
- Gated Check-ins
- Private Builds
- Properties for common customizations
- Integration with Symbol and Source server
- Advanced build delete options
- Build-related security permissions
- Build notifications



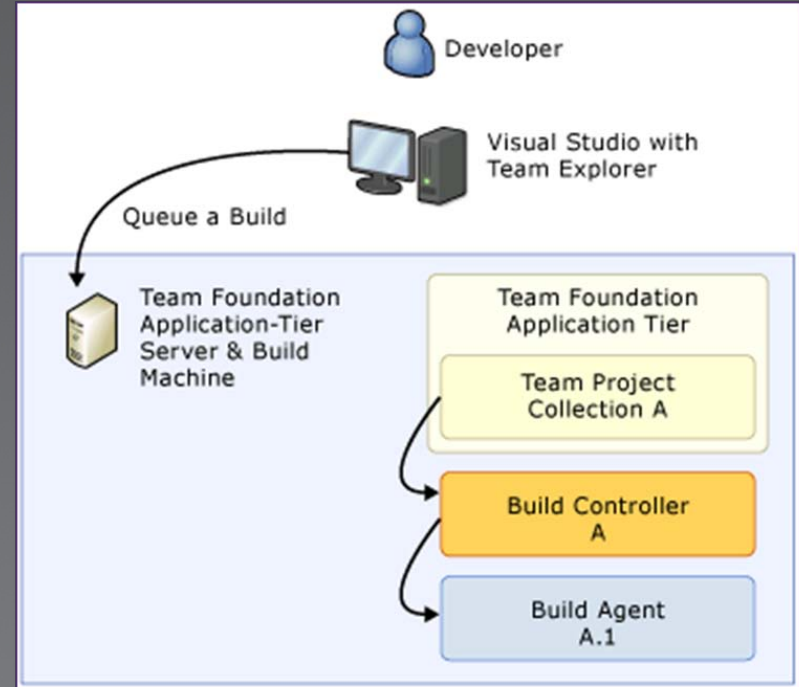
Build Configuration

- To use Team Foundation Build, you must have at least one build machine
 - Medium and large efforts will probably need multiple build machines
 - Build machines can be physical or virtual
- Components
 - Team Foundation Build Service
 - Build Controller
 - Build Agent



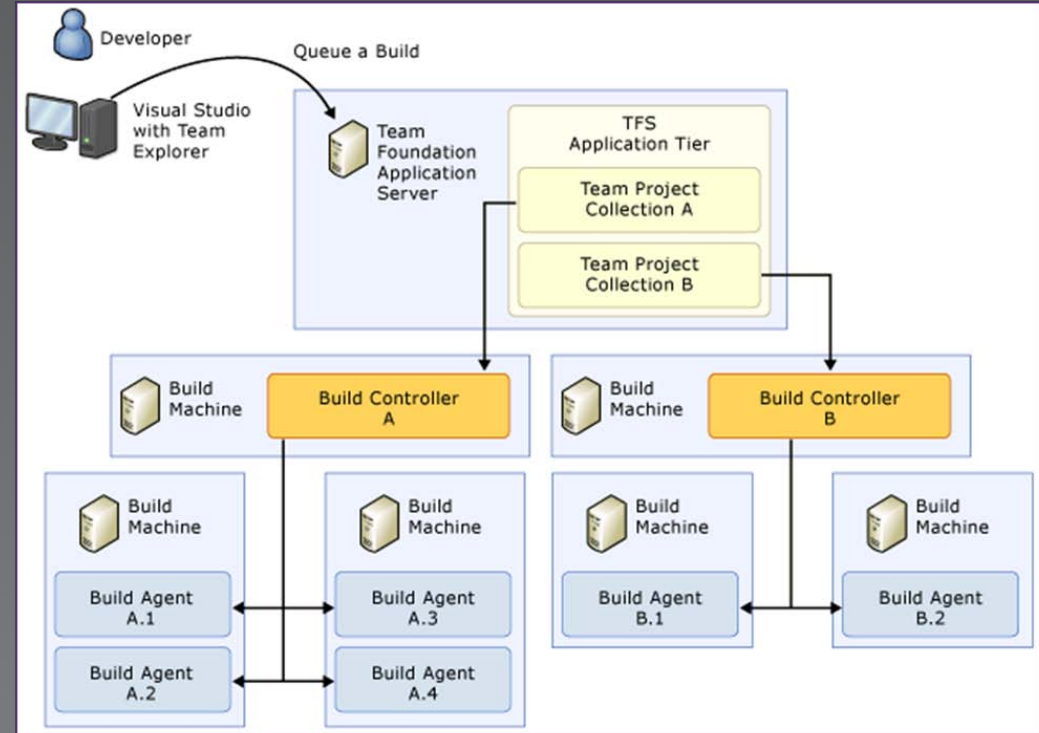
Simple Build Configuration

- Installed on same computer as App-Tier
- Supports less complex projects
- Can be expanded incrementally
 - Add additional Agents on separate machine(s)
 - Put the Build Controller on separate machine



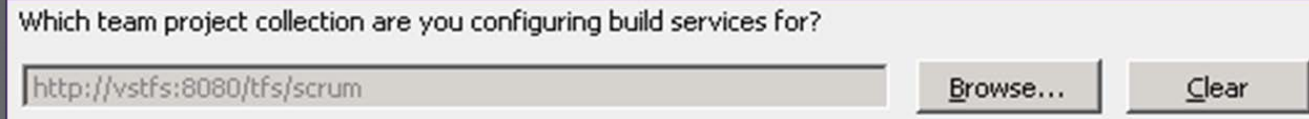
Enterprise-Level Build Configuration

- Multiple Team Project Collections
- Each has their own build controller, multiple build machines and agents



FAQ: Can one build machine support multiple TPCs?

- No, each build machine serves one TPC

A screenshot of a web-based configuration dialog box. The title bar reads "Which team project collection are you configuring build services for?". Below the title is a text input field containing the URL "http://vstfs:8080/tfs/scrum". To the right of the input field are two buttons: "Browse..." and "Clear".

Which team project collection are you configuring build services for?

- Organizations with multiple TPCs must install additional build machines
- For demonstration or testing purposes Jim Lamb posted a hack:
 - <http://bit.ly/bFbSsz>



FAQ: Will my 2008 build agents work with TFS 2010?

- No, you will need to install and configure Team Foundation Build 2010



FAQ: Can I build my VS 2010 solutions in TFS 2008?

- Yes, the TFS 2008 build agent can use MSBuild 4.0 to build VS 2010 solutions
- On the Team Foundation Build 2008 machine:
 - Install .NET 4.0
 - Install Visual Studio 2010 (if you are running tests)
 - Edit the TFSBuildService.exe.config file and set the MSBuildPath value to v4

```
<!-- MSBuildPath
    Set this value to the full path to the directory of MSBuild.exe to use
    a location other than the default. This should only be needed if a new
    version of the .NET Framework is installed.
-->
<add key="MSBuildPath" value="C:\Windows\Microsoft.NET\Framework\v4.0.30319" />
```

TFSBuild.exe Command-Line Tool

- **Start** a new build
 - Use with the Windows Task Scheduler service for more granular scheduling
- **Stop** a running build
- **Delete** a completed build
 - Details, DropLocation, Label, TestResults, Symbols, All
 - Nothing is deleted from the data warehouse
- **Destroy** a completed build
 - Destroys (purges) previously detailed builds



TFSBuild.exe Example

Syntax for starting a build

```
TFSBuild.exe start /collection:teamProjectCollectionUrl  
/builddefinition:definitionSpec  
[/droplocation:droplocation] [/getoption:getoption]  
[/customgetversion:versionSpec] [/priority:priority]  
[/requestedfor:userName] [/msbuildarguments:args]  
[/shelveset:shelvesetName [/checkin]]  
[/silent] [/queue]
```

Example

```
TFSBuild.exe start http://myserver:8080/DefaultCollection  
/builddefinition:"AdventureWorks\Nightly"  
/droplocation:"\\computername\buildDrops"
```


Build-Level Permissions (1 of 2)

Permissions		Groups			
Name	Description	Proj Admin	Builder	Contrib.	Reader
View builds	Can view the queued and completed builds for this team project	✓	✓	✓	✓
Edit build quality	Can add information about the quality of the build through the user interface for Team Foundation Build	✓	✓	✓	
Retain indefinitely	Can mark a build so that it will not be automatically deleted by any applicable retention policy	✓	✓		
Delete builds	Can delete a completed build	✓	✓		
Manage build qualities	Can add or remove build qualities	✓	✓		
Destroy builds	Can permanently delete a completed build	✓	✓		
Update build information	Can add information about the quality of a build	<i>Service Accounts Only</i>			

Build-Level Permissions (2 of 2)

Permissions		Groups			
Name	Description	Proj Admin	Builder	Contrib.	Reader
Queue build	Can put a build in the queue through the interface for Team Foundation Build or at a command prompt	✓	✓	✓	
Manage build queue	Can cancel, re-prioritize, or postpone queued builds	✓	✓		
Stop builds	Can stop a build that is in progress	✓	✓		
View build definition	Can view the build definitions that have been created for the team project	✓	✓	✓	✓
Edit build definition	Can create and modify build definitions for this project	✓	✓		
Delete build definition	Can delete build definitions for this project	✓	✓		
Override check-in validation by build	Can commit a changeset that affects a gated build definition without triggering the system to shelve and build their changes first	<i>Service Accounts Only</i>			

Team Foundation Build Components

DEMO

Build Definitions in Team Foundation Server 2010

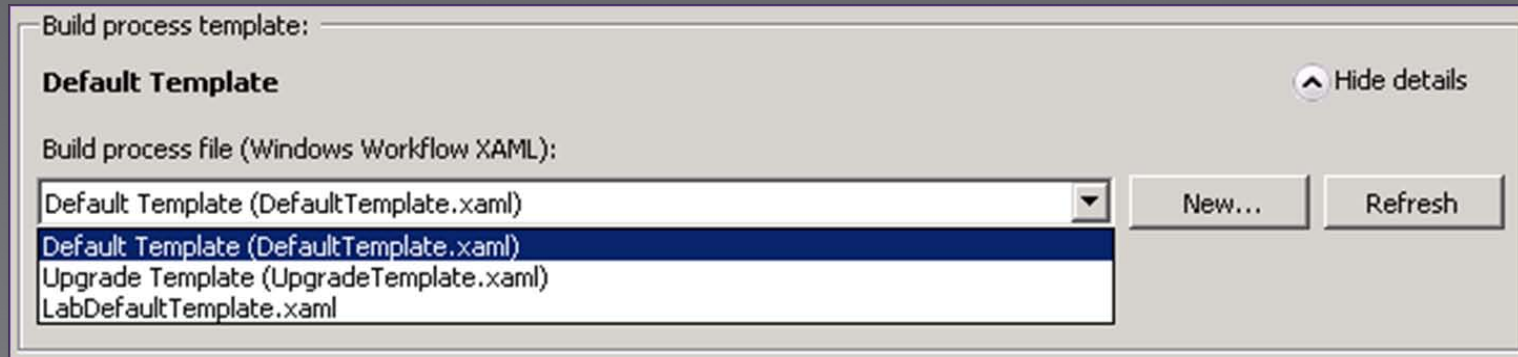
- TFS 2005 and 2008 builds were executed by MSBuild
- TFS 2010 builds primarily use Windows Workflow Foundation (WF) 4.0

	TFS 2005	TFS 2008	TFS 2010
Orchestration	MSBuild 2.0	MSBuild 3.5	WF 4.0
Building	MSBuild 2.0	MSBuild 3.5	MSBuild 4.0
Testing	MSBuild 2.0	MSBuild 3.5	WF 4.0
Deploying	MSBuild 2.0	MSBuild 3.5	WF 4.0



Build Process Templates

- WF 4.0 templates are used to define your build process
 - TFSBuild.proj files don't drive the build process anymore
- When defining a build, you can select a template or create a new, custom one



Upgrade Template (*UpgradeTemplate.xaml*)

- Allows you to continue to use TFSBuild.proj files
- When you upgrade to TFS 2010 your existing builds are converted to use this
 - Also, if you check-in a TFSBuild.proj file into a new folder under the TeamBuildTypes folder this will create a new build definition
- You can even take advantage of gated check-in and shelveset builds
- You may want to convert to WF-based builds to increase ease-of-use



FAQ: Did TFS 2010 fix the MAX_PATH problem when building?

max_path (măk'spăth) *n. pl.*

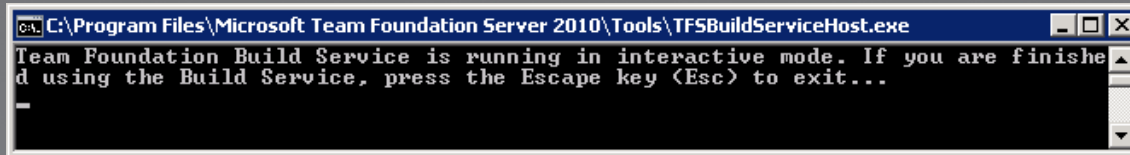
1. The maximum length of a path according to the Windows API, defined as 260 characters.
2. A royal PITA

- **Not really**, although Microsoft did change the default working folder to C:\Builds (rather than under the build service account's temporary directory like in 2008) and the location for a particular build definition was shortened in order to reduce the overall length

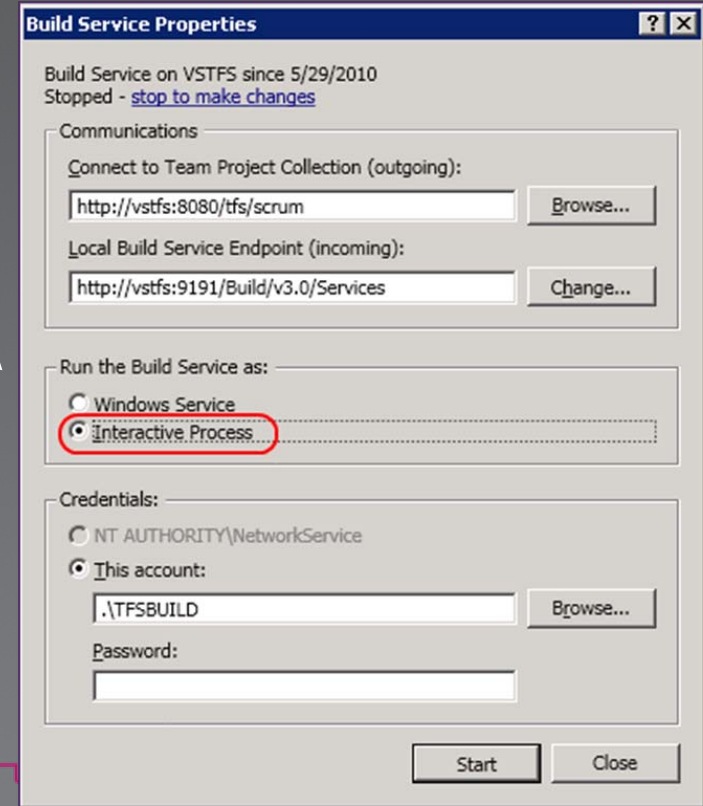


FAQ: How do I configure Team Build to run Coded UI Tests?

- You need to configure the Build Service to run as an Interactive Process
 - This is not the default value
 - You must use the TFS Admin Console
 - You will need to stop the build service first
- When you start the service, it will pop open a command window and wait for you to queue and execute a build



```
C:\Program Files\Microsoft Team Foundation Server 2010\Tools\TFSBuildServiceHost.exe
Team Foundation Build Service is running in interactive mode. If you are finished using the Build Service, press the Escape key <Esc> to exit...
-
```



Build Service Properties

Build Service on VSTFS since 5/29/2010
Stopped - [stop to make changes](#)

Communications

Connect to Team Project Collection (outgoing):
 [Browse...](#)

Local Build Service Endpoint (incoming):
 [Change...](#)

Run the Build Service as:

☐ Windows Service

☒ **Interactive Process**

Credentials:

☐ NT AUTHORITY\NetworkService

☒ This account:
 [Browse...](#)

Password:

[Start](#) [Close](#)

Workflow-Based Build Process

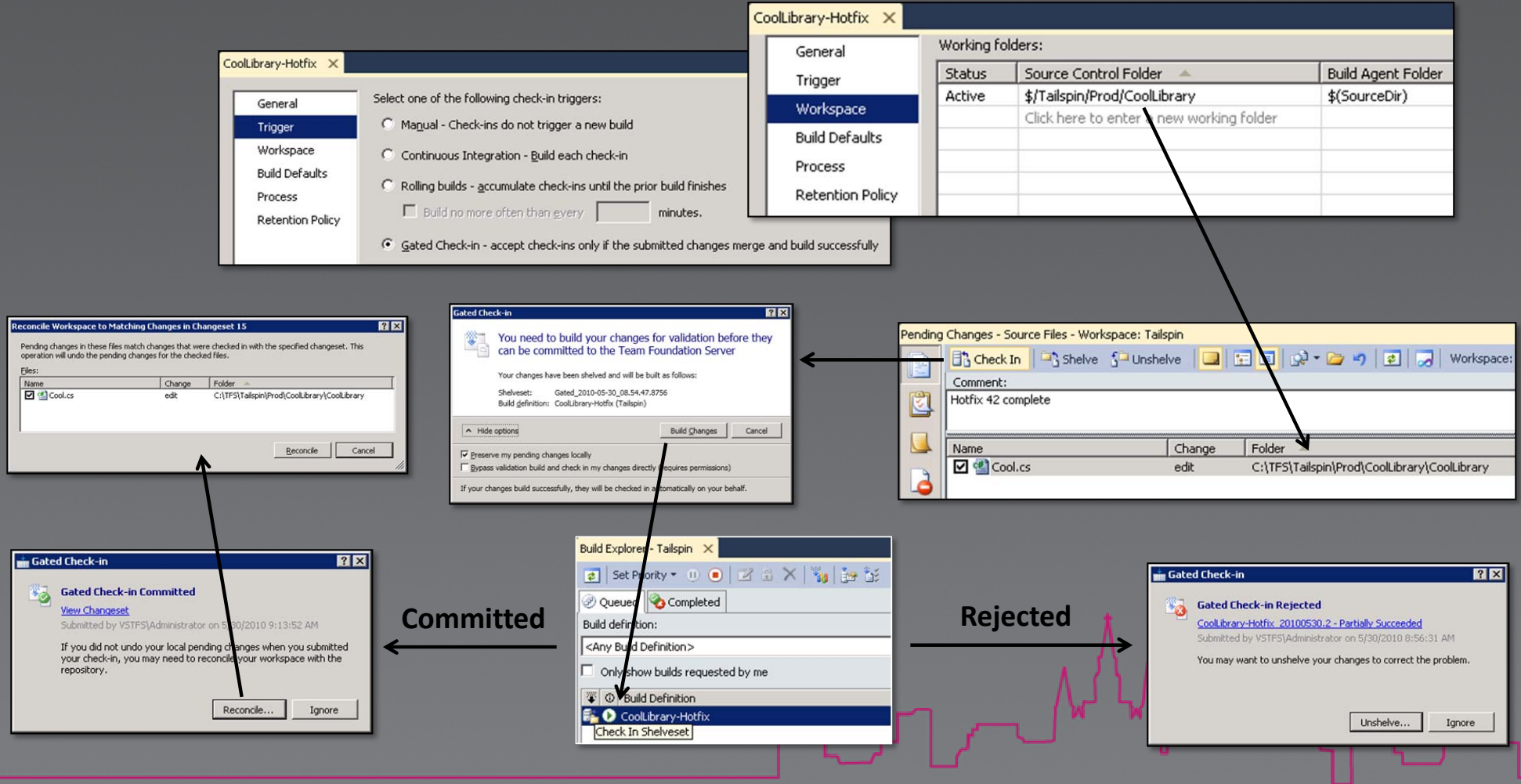
DEMO

Gated Check-in

- It's important to ensure that the latest code in your repository always builds without error and passes any associated automated tests
 - Agile teams can use Continuous Integration to achieve this
 - Larger teams may have difficulty accomplishing this
- Gated Check-ins address this problem
 - Pending changes are shelved by the developer
 - The build server takes the latest code and merges the shelve set code
 - If the code builds and the tests pass, code is checked-in (on behalf of the build server)



Gated Check-in Process



Private Builds (a.k.a. “Buddy Builds”)

- A Private build is just like a Gated check-in build, but without the automatic check-in and enforcement
 - It's a way to manually queue a Gated check-in build and provide a Shelveset name
 - It provides another sanity check beyond F5
- You specify a private build by selecting a Shelveset to merge with the latest source code when queuing a build

Queue Build "Tailspin"

General Parameters

Build definition:
CoolLibrary-Nightly

CoolLibrary Nightly Build

What do you want to build?
Latest sources with shelveset

Shelveset name:
Gated_2010-05-30_09.06.21.3662 ... Create...

☐ Check in changes after successful build

Build controller:
VSTFS - Controller

Priority in queue: Normal Position: 1

Drop folder for this build:
\\vstfs\BuildDrop

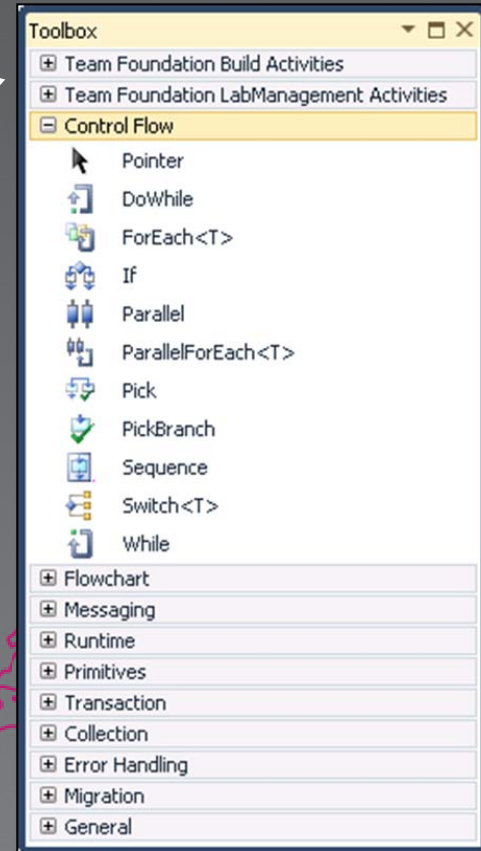
Queue Cancel

Gated / Private Builds

DEMO

Customizing Team Foundation Build 2010

- Team Build can be customized in many ways
 - Create a custom build process template
 - Add standard activities to your custom definition
 - Create custom activities and add them to your definition
- Tip: make sure there's not an out-of-the-box way of solving your problem first such as overriding a default behavior by setting a build parameter



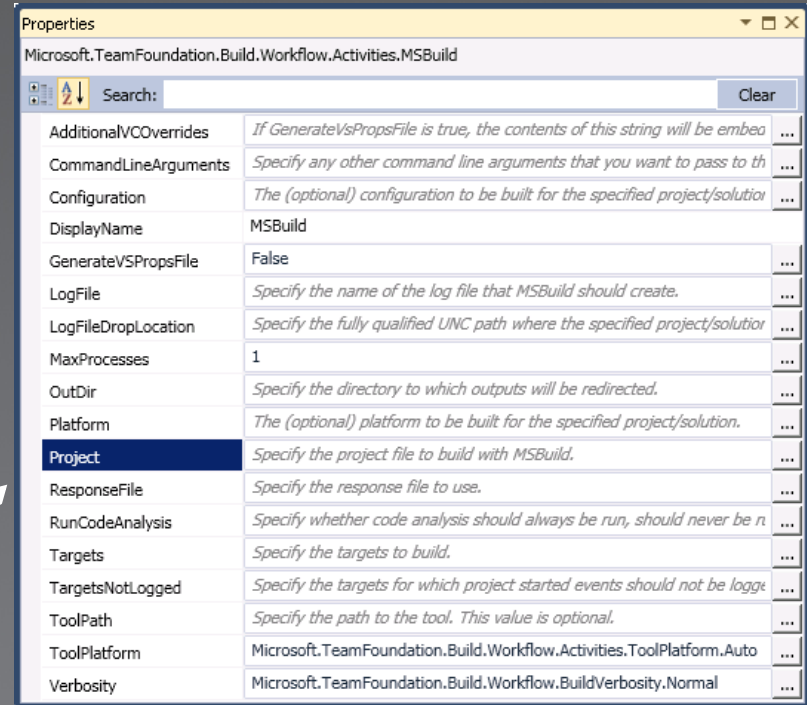
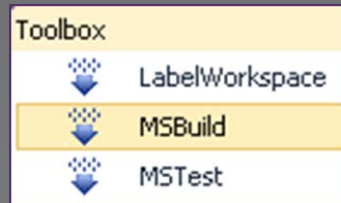
Build Process Parameters Alleviate Need for Customization

- Many customizations in 2005/2008 are now parameterized settings
- Parameter defaults can be changed on the build definition *after creation*

Property	Property	Property
Build individual .NET projects	Logging verbosity	Create work item on failure
Fail build on test failure	Perform Code Analysis	Disable tests
Test category criteria	Index Sources	Get version
Test priority filters	Publish symbols	Label sources
Run all tests in a .VSMDI file	Analyze Test Impact	MSBuild arguments
Build number format	Associate changesets & WIs	MSBuild Platform
Clean workspace	Copy Outputs to drop folder	Private drop location

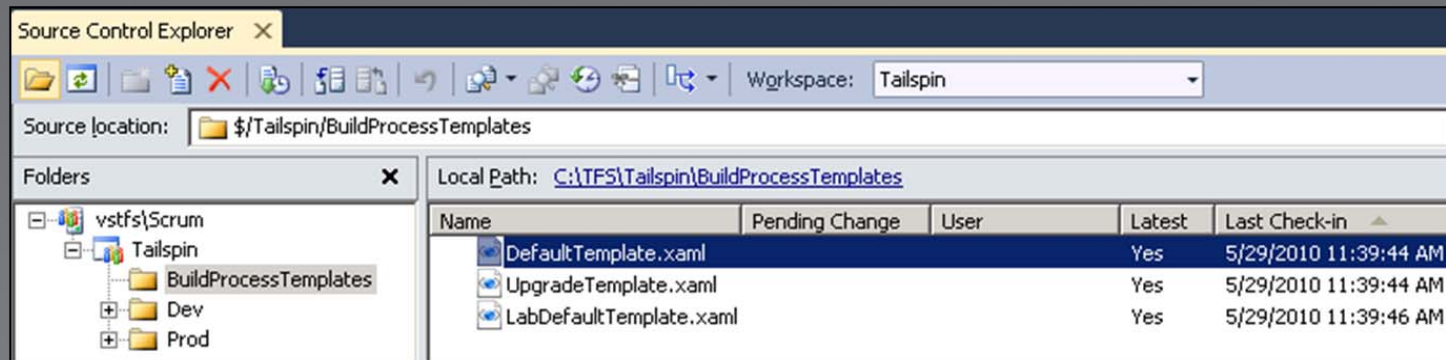
FAQ: Can I keep using my custom MSBuild tasks?

- Yes, if you are using *UpgradeTemplate.xaml*
- Maybe, if you are using *DefaultTemplate.xaml*
 - Yes, for the compiling phase if you configure the .proj file to call them
 - Yes, for other activities if you create a custom .proj file and then call it using the MSBuild activity



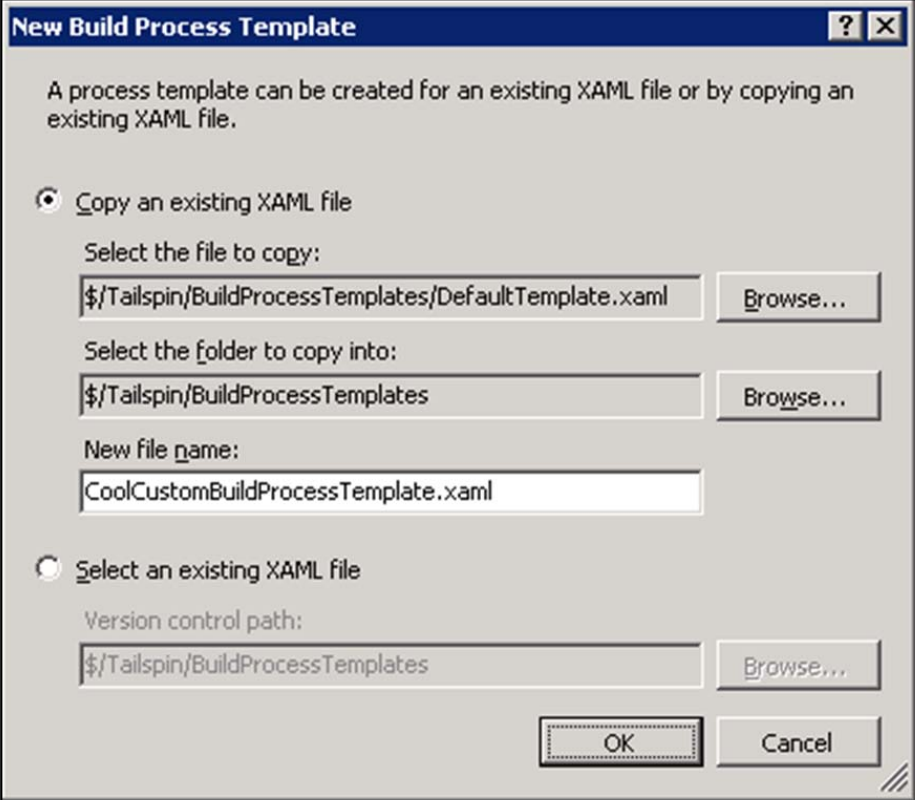
FAQ: Can I customize the out-of-the-box templates?

- Yes, but you should back them up or branch them first
- You should also plan on reviewing any build definitions that are based on the template and make any necessary fixes
- You should be safe when installing TFS service packs down the road because Microsoft typically doesn't service the contents of team projects and the worst case is you might need to manually merge some XML changes



Adding a New Build Process Template

- You can create a copy of an existing template
- You can copy/create a template manually and upload it to Team Foundation Server



The screenshot shows a dialog box titled "New Build Process Template". It contains two radio buttons for selection. The first radio button, "Copy an existing XAML file", is selected. Below it are three text boxes: "Select the file to copy:" with the path "\$/Tailspin/BuildProcessTemplates/DefaultTemplate.xaml", "Select the folder to copy into:" with the path "\$/Tailspin/BuildProcessTemplates", and "New file name:" with the text "CoolCustomBuildProcessTemplate.xaml". Each text box has a "Browse..." button to its right. The second radio button, "Select an existing XAML file", is unselected. Below it is a text box for "Version control path:" with the path "\$/Tailspin/BuildProcessTemplates" and a "Browse..." button to its right. At the bottom right are "OK" and "Cancel" buttons.

New Build Process Template

A process template can be created for an existing XAML file or by copying an existing XAML file.

☒ **C**opy an existing XAML file

Select the file to copy:

\$/Tailspin/BuildProcessTemplates/DefaultTemplate.xaml **B**rowse...

Select the folder to copy into:

\$/Tailspin/BuildProcessTemplates **B**rowse...

New file name:

CoolCustomBuildProcessTemplate.xaml

☐ **S**elect an existing XAML file

Version control path:

\$/Tailspin/BuildProcessTemplates **B**rowse...

OK **Cancel**

FAQ: Can I delete a build process template?

- Yes, but make sure you first locate and delete all build definitions that are based on it
- When you delete the template, any remaining build definitions based on the template will no longer function



Call an MSBuild Task by Using the MSBuild Activity

- Create an MSBuild project with a single target that calls the task
- Use the MSBuild WF Activity to call the MSBuild project

Pros	Cons
Simple	Offers a limited debugging experience
No coding is required	Offers limited testability
Leverages existing MSBuild knowledge	You won't have full access to the WF runtime
No need to modify the MSBuild task	Not a simple drag and drop experience
Not much WF knowledge is required	May need to create a custom WF activity to properly handle input and output properties
The task will still be useable by MSBuild	

Create a Custom WF Activity

- Can be implemented in code or in XAML
 - Create a C# Activity Library to hold both the XAML and the code activity
 - Specify any variables and arguments
 - Unit test your code!
- Check-in the assembly and tell the Build Controller where it lives

Pros	Cons
Easy to handle output and complex properties	Coding is required
Improved debugging experience	Knowledge of WF is required
Can take advantage of the WF runtime	Not callable by MSBuild (without extra work)
Simple drag and drop experience	
Minimal ongoing maintenance	

Customizing the Build Process

DEMO

Professional Scrum Developer Program



- An innovative program for developers from Microsoft and the founders of Scrum
- *Learn how to use modern engineering practices to develop an increment of complete, potentially shippable functionality using Visual Studio 2010, ALM, and the Scrum framework*
- Training course, assessment, and certification available
- Visit MSDN for more details: <http://bit.ly/dppXd0>

ANNOUNCING

Resources



Sessions On-Demand & Community

www.microsoft.com/teched



Microsoft Certification & Training Resources

www.microsoft.com/learning



Resources for IT Professionals

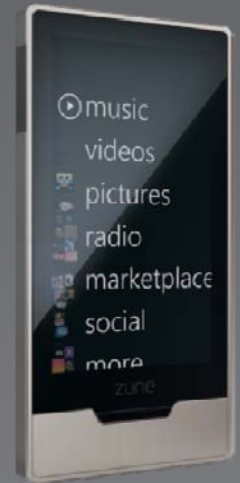
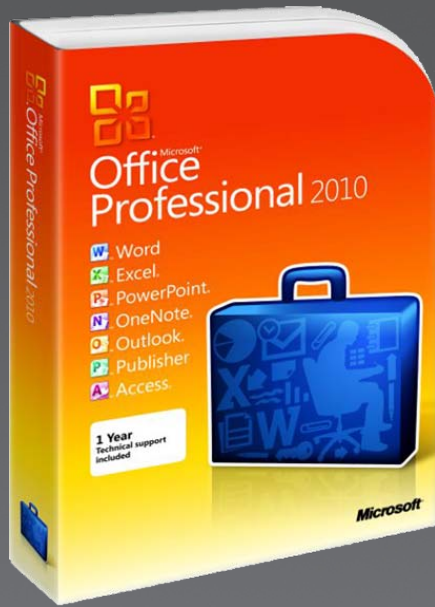
<http://microsoft.com/technet>



Resources for Developers

<http://microsoft.com/msdn>





Complete an
evaluation on
CommNet and
enter to win!

Microsoft®

© 2010 Microsoft Corporation. All rights reserved. Microsoft, Windows, Windows Vista and other product names are or may be registered trademarks and/or trademarks in the U.S. and/or other countries.
The information herein is for informational purposes only and represents the current view of Microsoft Corporation as of the date of this presentation. Because Microsoft must respond to changing market conditions, it should not be interpreted to be a commitment on the part of Microsoft, and Microsoft cannot guarantee the accuracy of any information provided after the date of this presentation.
MICROSOFT MAKES NO WARRANTIES, EXPRESS, IMPLIED OR STATUTORY, AS TO THE INFORMATION IN THIS PRESENTATION.

Related Content

- Breakout Sessions (session codes and titles)
- Interactive Sessions (session codes and titles)
- Hands-on Labs (session codes and titles)
- Product Demo Stations (demo station title and location)



Track Resources

- Resource 1

- Resource 2

- Resource 3

- Resource 4

